



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS : X
SUBJECT: MATHEMATICS

FULL MARKS: 40
DATE: 11.10.22

General Instructions:

- The paper consists of three printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION: A

1. i) $(\sec A + \tan A)(1 - \sin A) =$ [8 × 1 = 8]

- a) $\sec A$ b) $\sin A$ c) $\operatorname{cosec} A$ d) $\cos A$

ii) A child has a die whose six faces show the letters as given below:

A	B	C	D	E	F
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The die is thrown once. The probability of getting a 'C' or 'D' is

- a) $1/2$ b) $1/6$ c) $1/3$ d) $1/4$

iii) The tops of two poles of height 20 m and 14 m are connected by a wire. If the wire makes an angle of 30° with horizontal, the length of the wire is

- (a) 12 m (b) 10 m (c) 18 m (d) 20 m

iv) For what value of x, the inclination of the line passing through the points P(3, 4) and Q(x, 5) is 45° ?

- a) 5 b) 4 c) 3 d) 2

v) The mean of 16 numbers is 8. If 2 is added to every number, what will be the new mean?

- a) 7 b) 9 c) 10 d) 11

vi) If A(5, -1), B(-3, -2) and C(-1, 8) are the vertices of triangle ABC, the length of the median through A is:

- a) $\sqrt{35}$ units b) $\sqrt{45}$ units c) $\sqrt{55}$ units d) $\sqrt{65}$ units

vii) The midpoint of the line segment joining (2a, 4) and (-2, 2b) is (1, 2a + 1). The values of a and b are :

- a) a = 1, b = 2 b) a = 2, b = 3 c) a = 3, b = 2 d) a = 2, b = 1

viii) The numbers 3 and 6 have frequencies $(y + 1)$ and y respectively. If their arithmetic mean is 4, then the value of y is:

- a) 5 b) 3 c) 1 d) -3

SECTION: B

2. Prove the following identities:

a) $\frac{\sec \theta - \tan \theta}{\sec \theta + \tan \theta} = 1 - 2 \sec \theta \tan \theta + 2 \tan^2 \theta$

b) $\frac{\sin \theta + 1 - \cos \theta}{\sin \theta - 1 + \cos \theta} = \frac{1 + \sin \theta}{\cos \theta}$

[4+ 4= 8]

3.a) The following table gives the life time of 400 LED bulbs:

Life time (in hours)	300-400	400-500	500-600	600-700	700-800	800-900	900-1000
Number of lamps	14	56	60	86	74	62	48

A bulb is selected at random. Find the probability that the life time of the selected bulb is:

- i) less than 400 hours
- ii) between 300 and 800 hours
- iii) atleast 700 hours.

[4]

4. The following table gives the daily wages of 50 workers of a factory. Calculate the mean daily wage of a worker of the factory by the short - cut method.

Wages(in ₹)	120- 130	130- 140	140- 150	150-160	160-170	170-180	180-190
No of workers	2	3	5	7	21	10	2

[4]

5. The line $x - 4y - 6 = 0$ is the perpendicular bisector of the line segment PQ and the coordinates of P are (1, 3), find the coordinates of Q. [4]

6. Given that the mean of the following distribution is 30, find the missing frequency “f”.

Class Interval	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
Frequency	4	6	10	f	6	4

[4]

7. A straight line with a negative gradient passes through P (2, 1) and cuts the X axis at A and Y axis at B. If BP: PA = 3: 1. Find:

i) the coordinates of A and B

ii) the equation of the line AB

[4]

8. A drone camera is used to shoot an object P, on the ground, from two different positions R and S along the same vertical line QRS, where Q is on the ground. The angle of depression of the object P from the two positions R and S is 35° and 60° respectively. If the distance of the object P from point Q is 50 metres, then find the distance between R and S correct to the nearest metre.

[4]